

LARGE VOLUME PARACENTESIS — Procedure One-Pager

1 · INDICATIONS

Therapeutic (LVP)	Symptomatic ascites: distention/pain, early satiety , dyspnea. (Asymptomatic large ascites alone is NOT an indication.)
Diagnostic	New-onset ascites; EVERY cirrhotic admitted with ascites (r/o SBP), even if asymptomatic

2 · CONTRAINDICATIONS

Defer LVP	Hemodynamic instability / sepsis (large-volume removal worsens circulatory dysfn); surgical abdomen; DIC / clinical fibrinolysis
NOT contraindic.	INR >2, plt <50K (no routine FFP/plt); intra-abd HTN (relieved BY tap). Pregnancy/prior surgery = use US, avoid scars

Coags don't block the tap: paracentesis is safe across a wide INR/plt range; AASLD recommends against routine correction. It's the hemodynamics that defer a *large-volume* tap.

3 · US SITE SELECTION

- **Low-freq curvilinear/phased-array:** safe pocket **>2–3 cm** (peritoneum → viscera, all 3 dim); **>5 cm** to justify a therapeutic tap. Note probe angle = needle angle.
- **Site:** LLQ/RLQ **~2 cm sup + ~2 cm medial to ASIS**, lateral to rectus.
- **High-freq linear + Doppler:** screen superficial abd-wall veins (light pressure) + identify **inferior epigastric vessels** (lateral to rectus).

SAFETY: stay LATERAL to the rectus — the **inferior epigastric arteries** run just lateral to the rectus sheath; a too-medial stick → rectus-sheath hematoma.

4 · TECHNIQUE

- **Time-out** w/ RN before opening sterile supplies.
- **Gear:** mask, eye protn, sterile gloves (cap/gown optional). Chlorhexidine **scrub ×2, 30 s** (iodine = apply & dry).
- **Anesthesia:** 1% lido wheal → 21 g under negative pressure to fluid → ~2 cc at peritoneum.
- #11 nick → catheter-over-needle (**Z-track**) under negative pressure to fluid → advance 0.5 cm, thread catheter to hub, remove needle.
- **Drain:** simple IV tubing (**NO valves**) to vacuum bottle; exchange until flow stops + symptoms improve.

5 · ALBUMIN

6–8 g of 25% albumin per liter removed for taps **>5 L** (~75 g for 10 L). Prevents post-paracentesis circulatory dysfunction; improves survival (AASLD 2021). Not needed for <4–5 L.

6 · FLUID STUDIES

Suspect	Send
Every tap	Cell count + diff; on 1st tap albumin + total protein (SAAG)
Infection (SBP)	Gram stain + culture (bedside into blood-cx bottles). PMN ≥250 = SBP → ceftriaxone
Malignancy	Cytology

High SAAG (≥1.1) + HIGH protein (>2.5) → cardiac/post-hepatic cause → **echocardiogram** (cirrhotic ascites is high-SAAG, LOW-protein).

FLOW STOPPED? — TROUBLESHOOT

- Turn off suction → **reposition** (raise HOB / roll to side)
- **Flush** catheter 5–10 cc saline (bowel plugging it)
- **Rotate** catheter
- **Withdraw** to ~2 cm + skin-to-peritoneum depth (~4 cm)
- Too little fluid (<2 cm) → dynamic US or IR

MASTERY-LEARNING THRESHOLDS

Component	MPS (sim)	UPS (clinical)
Mistakes /41 (Angoff)	<30	<8
Global skills	>2	≥proficient
Entrustment	>2	w/o direct supervision